C2.2 Neural signalling

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-c2-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 479–489 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

ACTION POTENTIALS

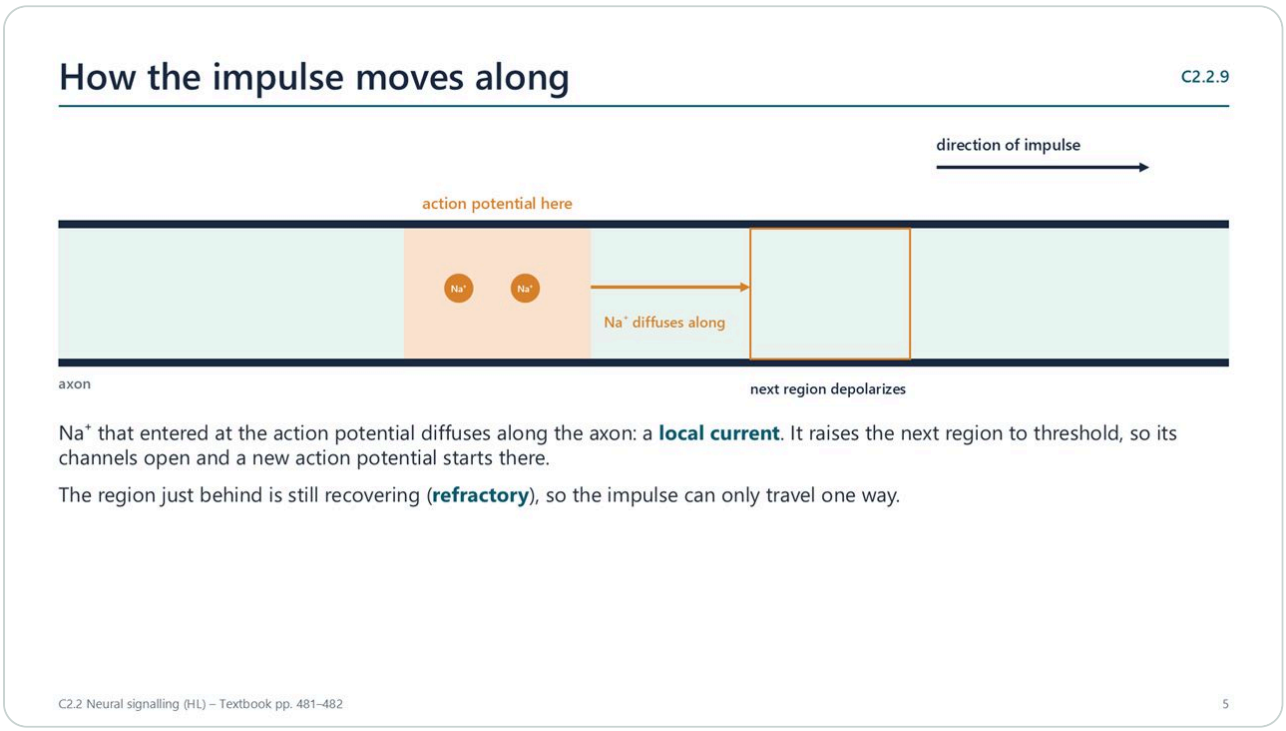
C2.2.8 Depolarization and repolarization

pp. 479–481

When the membrane reaches the **threshold** (about −50 mV), voltage-gated Na^+ channels open and Na^+ diffuses in: the inside becomes positive (**depolarization**). At about +30 mV they close and voltage-gated K^+ channels open; K^+ diffuses out and the inside becomes negative again (**repolarization**). Below threshold, no action potential occurs: it is all-or-nothing.

C2.2.9 Local currents

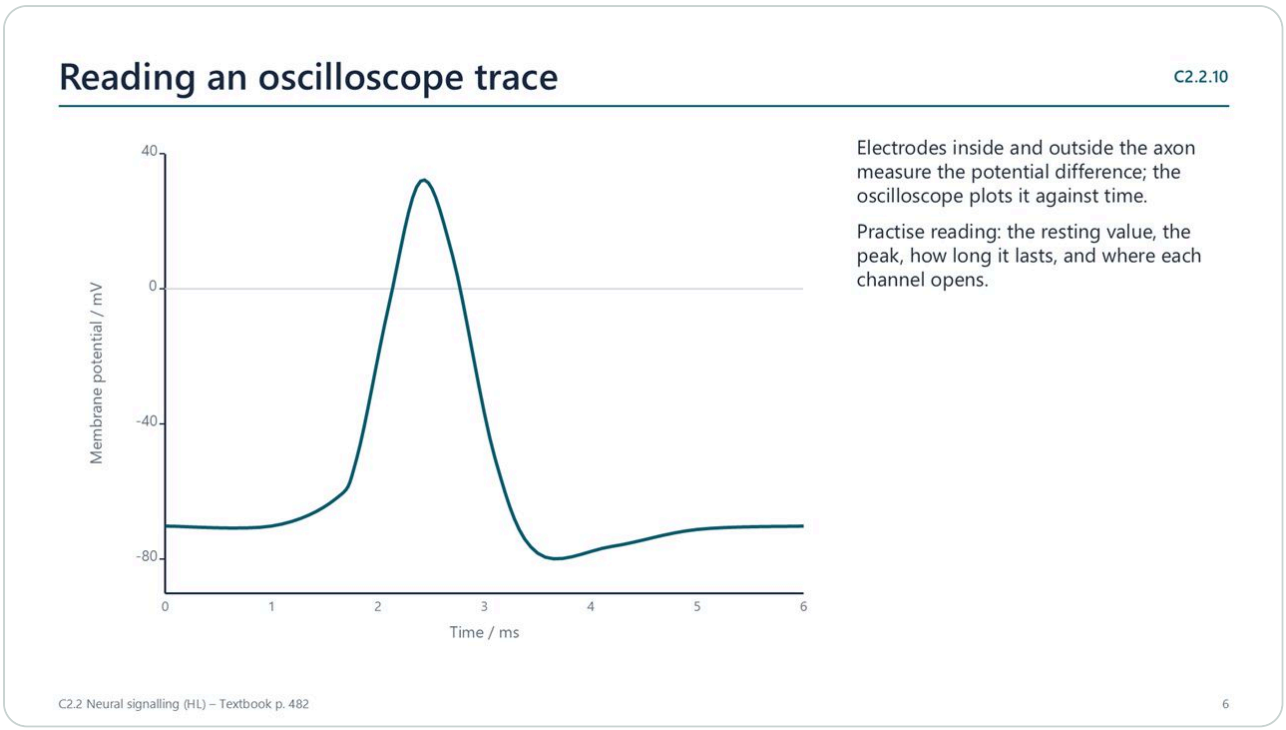
pp. 481–482



Na^+ that has entered diffuses along inside the axon, while Na^+ outside diffuses towards the depolarized region. These local currents bring the next region to threshold, so its Na^+ channels open and the impulse moves on.

C2.2.10 Oscilloscope traces

p. 482



An oscilloscope plots membrane potential against time: a flat resting potential near −70 mV, a steep rise to about +30 mV, a fall, a brief dip below rest, then recovery, all within a few milliseconds.

C2.2.11 Saltatory conduction

pp. 482–483

Myelin insulates the axon. Pumps and channels are clustered at the **nodes of Ranvier**, so the action potential jumps from node to node. Myelinated axons conduct many times faster than unmyelinated ones of the same width.

SYNAPSES

C2.2.12 Exogenous chemicals

pp. 483–484

Neonicotinoid pesticides bind nicotinic acetylcholine receptors in insects and are not broken down, so transmission is blocked. **Cocaine** blocks the transporters that reabsorb dopamine, so dopamine stays in the cleft and keeps stimulating.

C2.2.13 Inhibitory neurotransmitters

pp. 484–485

Neurotransmitters such as GABA open channels that let chloride ions in or potassium ions out. The postsynaptic membrane becomes **hyperpolarized**, an inhibitory postsynaptic potential, so an action potential is less likely.

C2.2.14 Summation

p. 485

A neuron receives many excitatory and inhibitory inputs. Their effects add together; only if the total reaches threshold does the neuron fire, and then fully.

PAIN AND CONSCIOUSNESS

C2.2.15 Pain

pp. 486–487

Free nerve endings in the skin have channels for positive ions that open with heat, acid or chemicals such as capsaicin. Ions enter, threshold is reached and impulses travel to the brain, where pain is perceived.

C2.2.16 Consciousness

pp. 487–489

Consciousness is an **emergent property**: it is not found in any single neuron but arises from the interactions of billions of neurons in the brain.

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